AMD DEVELOPER HACKATHON · TRACK 1

TokenGolf

Track 1 ranks by **fewest tokens**. The catch: the only allowed models are **reasoning models** that burn hundreds of hidden tokens per answer. We turned that off. **–46% scored tokens, higher accuracy.**

$ Fireworks AI · llama.cpp · Docker linux/amd64

fewest strokes wins

$ THE GAME

Clear an accuracy gate, then rank by fewest Fireworks tokens. The only lever that helps is spending less.

THE TRAP

Both allowed models, minimax-m3 and kimi-k2p7-code, are **reasoning models**. They think out loud in hidden tokens before answering, and every token is scored.

THE COST

A one-word sentiment label cost **41–68 output tokens** of invisible reasoning. Across eight categories, that hidden thinking was our single biggest scored cost.

Everyone pays the accuracy gate. The rank is won on tokens nobody sees.

$ THE STROKE

One flag flips it: reasoning_effort = "none".

sentiment "positive" 41-68 tok → 2 tok

kimi mean / answer 341 tok → 185 tok (-46%)

accuracy .917 → .958 (went up)

Measured, not assumed. "low" was ignored by these models; chat_template_kwargs{enable_thinking:false} returned 400.
"none" is the only lever that fires, and it costs no accuracy.

$ HOW IT PLAYS

Read the task, take one shot, write the answer.

01 **Read `/input/tasks.json`**

Eight categories: factual QA, math, sentiment, summarisation, NER, code-debug, logic, code-gen.

02 **Route: free local first, else one Fireworks call**

Optional local tier (Qwen2.5-3B on CPU, **0 scored tokens**) answers sentiment & NER. Everything else takes a single Fireworks call with reasoning off.

03 **Binary cascade, never a chain**

Because every call adds tokens, we never escalate model-to-model. Local → one remote call, terse and capped. That's it.

04 **Write `/output/results.json`**

Exact harness contract: every call routed via FIREWORKS_BASE_URL, models from ALLOWED_MODELS.

$ KNOWING WHEN TO ESCALATE

A calibrated gate spends a paid call only when it isn't sure.

FREE CONFIDENCE

The local model answers a few times; agreement across samples is a self-consistency confidence signal, and sampling on local costs nothing.

CALIBRATED, NOT GUESSED

That signal is calibrated on held-out data, so the escalation threshold is measured rather than hand-tuned. High confidence → keep the free answer; low → escalate.

The honest default when unsure is to **spend one token-cheap Fireworks call**, not bluff a free answer that fails the gate.

$ THE SCOREBOARD

Fewer tokens, and a higher gate score.

-46%	.958	259	8/8
mean scored tokens (kimi, 341 → 185)	accuracy with reasoning off (up from .917)	tokens on a live 3-task run (was 493, same answers)	categories handled through one contract

Ships as linux/amd64 · cold-start 1.7s (limit 60s) · slowest request 4.4s (limit 30s) · image < 10 GB · 199 tests green.

$ STATED HONESTLY

What we measured, and what we didn't overclaim.

- ✓ Accuracy numbers are **n=3 per category**. The token cut is robust and per-call verified; the accuracy edge is directional, not proven.
- ✓ Latency was **de-risked by proxy** on our machine, not the real judging VM. Comfortable even at 5× slower, but stated as a proxy.
- ✓ Grammar-constrained decoding (from a paper sweep) was **tested and reverted**: 0 fixed, 1 broken on our checkers. A measured negative, kept out.
- ✓ The free-local tier's edge **shrank** once reasoning was off, so we lead with the lean Fireworks image and treat local as optional.

Reason less, score more.

TokenGolf: a calibrated, token-efficient routing agent for AMD Track 1.

image	docker.io/soren19/router-agent:kimi
code	github.com/nguiaSoren/router-agent